Adam K. Cheong

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Inventive high school senior with a strong interest in biomechanical engineering and robotics; founding intern of Watney Robotics, a venture-backed startup; winner of multiple STEM awards and a Masason Scholar since 2025.

EDUCATION

2023–2027 Proof School, San Francisco, CA

AWARDS

Sept 2026 National Merit Commended Scholar
July 2026 AP Scholar with Distinction
March 2026 Honorable Merit, USA Biolympiad
Dec 2025 AIME Qualifier
Sept 2025 Ellison Scholar (Semi-Finalist)
July 2025 Masason Scholar (2025–2031)
April 2025 Cum Laude, National Latin Examination (Honorable Mention)
March 2025 Honorable Mention, USA Biolympiad
April 2024 Maxima Cum Laude, National Latin Examination (Silver Recognition)

STEM FAIRS

Feb 2027 Golden Gate STEM Fair (GGSF) 2027
Nov 2026 Grand Jamboree, International Genetically Engineered Machine (iGEM)
March 2025 Fourth Place, Computational Biology. High School Division, Alameda County Science & Engineering Fair (ACSEF)
April 2024 Honorable Mention, Zoology. Senior Division, California Science & Engineering Fair (CSEF)
March 2024 First Place, Biological Sciences. 9th Grade, Golden Gate STEM Fair (GGSF)
April 2023 Honorable Mention, Applied Mechanics & Structures. Junior Division, California Science & Engineering Fair (CSEF)
March 2023 First Place, Engineering: Electrical & Mechanical. 8th Grade, Golden Gate STEM Fair (GGSF)
March 2023 First Place, Engineering, Technology & Application of Science. 8th Grade, San Mateo County STEM Fair

EXPERIENCE

Watney Robotics, San Francisco, CA (2023–present)

- CAD modeling using Onshape; manage 3D printers
- Built server rack peripherals such as wire harness
- Assembled robot arm controller; trained ML models

VOLUNTEERING

- Student Ambassador, Berkeley Pre-College (‘26–27)
- Student Volunteer, 16th Gathering 4 Gartner (‘26)

SOCIETIES & CLUBS

- KLS iGEM Team @ BioCurious Lab (2026–present)
- Biomedical Engineering Society (2026–present)
- Genetics Society of America (2025–present)
- Dogpatch / Benchmark Climbing (2023–present)
- Berkeley Scuba Club (2025–present)
- Berkeley Math Circle (2015–2025)

SUMMER COURSES

- DATA C8, Berkeley Pre-College Scholar (Summer ’26)
- CS 61BL, Berkeley Pre-College Scholar (Summer ’26)
- CS 61A, Berkeley Pre-College Scholar (Summer ’25)
- Engineering & Robotics Workshop, MIT (Summer ’24)
- Game Theory, Johns Hopkins CTY (Summer ‘24)
- Principles of Logic, Johns Hopkins CTY (Summer ’23)

QUALIFICATIONS

- **Skills:** Use Onshape CAD to build server rack peripherals, prototype wire harness of robot controllers, assemble robot arm controllers, solder circuit boards, assemble PC, train ML models.
- **Languages:** Latin (2 years), Japanese (limited phrases), Mandarin (limited phrases), Spanish (conversational, 6 years), English (native speaker).
- **Assessments:** SAT 1550 (760 EBRW, 790 Math).

CERTIFICATIONS

- PADI Advanced Open Water Diver certification.
- Biosafety Level 1 (BSL-1) safety training.
- CPR, AED, and basic first aid certifications.

POSTERS

July 2027 How to Train Your C. Elegans: Inducing Thermotactic Behavior in a Digitally Simulated Worm. 26th International Worm Meeting, Kobe, Japan (submit abstract by March 10, 2027).

Oct 2026 How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions. 2026 Biomedical Engineering Society Annual Meeting, High School Poster Expo, Orlando, Florida.

June 2026 How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions. 2026 Population, Evolutionary & Quantitative Genetics Conference, Asilomar Conference Grounds, California.

June 2025 Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms. 25th International Worm Meeting, UC Davis, California.

Feb 2025 A Twist on Decussation. Junior Science & Humanities Symposium, San Francisco State University, California.

PAPERS

July 2027 How to Train Your *C. Elegans*: Inducing Thermotactic Behavior in a Digitally Simulated Worm. First draft of paper expected to be ready before 26th International Worm Meeting, July 2027.

Feb 2025 Why Do Vertebrates Have Decussated Corticospinal Tracts? Submitted to Junior Science & Humanities Symposium, NorCal, 2025.

PROJECTS

Aug–May 2027 An ongoing project to simulate biomechanical behaviors of a digital worm inside an artificial aquarium to study AFD-AIY connectivity along the worm's thermosensory pathway known to drive thermotaxis in response to a temperature gradient.

Jan–Nov 2026 A team project participated by students from 9 Bay Area high schools for the iGEM competition to engineer a biosensor that responds to rising levels of cadaverine, an organic compound from decomposing meat, by producing a blue chromoprotein.

Jan–Apr 2026 A joint proposal to study cancer cells population dynamics driven by tumor growth vs. dose response within a press-pulse framework in collaboration with university student researchers.

Jan–Apr 2025 A failed attempt to recreate evolutionary events inside a computer (software kept crashing); but would otherwise be interesting to see how a digital worm evolves *in silico* to become the 'first fish' after a somatic twist.

Jan–Mar 2024 Why do vertebrates have decussated corticospinal tracts? A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.

Jan–Mar 2023 A novel mountain climbing aid that brings a new mechanical advantage to mountaineers by means of a simple lever mechanism — inspired by a physics demo harnessing a setup of three toothpicks to carry a payload hanging off the edge of a table.

REFERENCES

- Dr. Michael Yetman
  Director of Science Programs
  Proof School
- Ryan Gannon
  Co-founder and CEO
  Watney Robotics